

# SmolVLA: Dual-Stream Flow Matching with Frozen Vision-Language Prio

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## Abstract

Building upon the foundational constraints of purely from-scratch models, SmolVLA introduces pre-trained knowledge distillation by leveraging a frozen OpenAI CLIP ViT-B/32 backbone. This dual-stream architecture processes raw language instructions and camera observations into a unified multimodal context via hierarchical cross-attention layers. The result is a highly parameter-efficient policy network that achieves robust zero-shot generalization while delegating continuous trajectory regression to a lightweight OT-CFM action expert.

## 1. Architectural Overview

SmolVLA is composed of two primary modules: the pretrained Multimodal Backbone and the Flow-Matching Action Decoder. The backbone utilizes frozen CLIP weights, ensuring that the vast web-scale pre-training is completely preserved without catastrophic forgetting. Images are tokenized into 49 patches, and instructions are parsed into 77 byte-pair encoded tokens. These independent embeddings are projected to a 128-dimensional latent space before fusion.

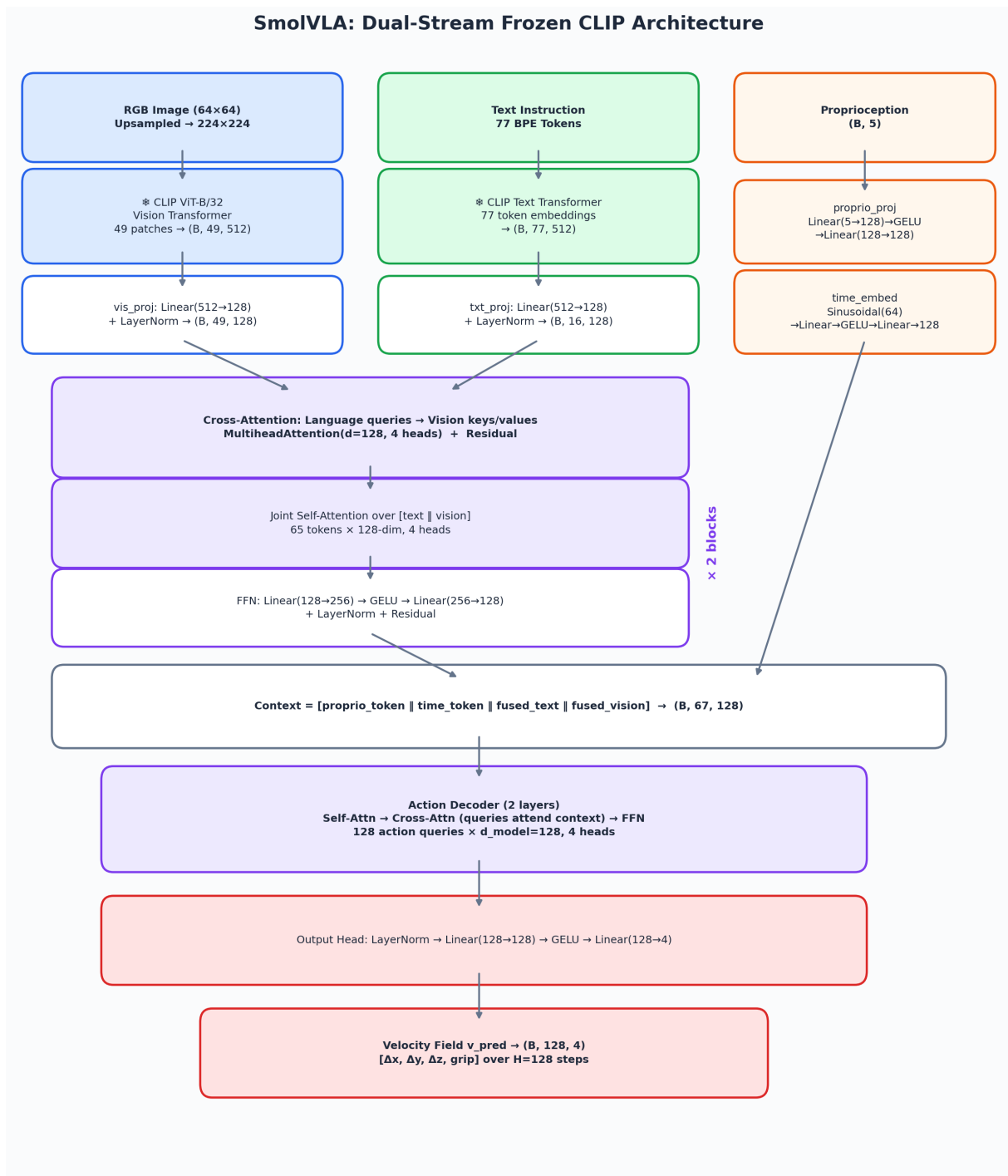


Figure 1: SmolVLA Dual-Stream Architecture integrating a frozen CLIP Backbone with Flow Matching.

## 1.1 Extracted Network Dimensions

To explicitly define the model complexity and tensor routing, we provide a live network summary extracted from the initialized PyTorch module. Note the parameter efficiency obtained by freezing the CLIP components.

| Layer (type:depth-idx)     | Output Shape | Param #       |
|----------------------------|--------------|---------------|
| SmolVLAPolicy              | [1, 128, 4]  | 16,384        |
| +SmolVLABackbone: 1-1      | --           | --            |
| +PretrainedVLMEncoder: 2-1 | --           | --            |
| +CLIP: 3-1                 | --           | (151,277,313) |
| +Sequential: 2-2           | [1, 16, 128] | --            |
| +Linear: 3-2               | [1, 16, 128] | 65,664        |

|                                         |  |                                      |               |         |
|-----------------------------------------|--|--------------------------------------|---------------|---------|
|                                         |  | +-LayerNorm: 3-3                     | [1, 16, 128]  | 256     |
|                                         |  | +-Sequential: 2-3                    | [1, 49, 128]  | --      |
|                                         |  | +-Linear: 3-4                        | [1, 49, 128]  | 65,664  |
|                                         |  | +-LayerNorm: 3-5                     | [1, 49, 128]  | 256     |
|                                         |  | +-ModuleList: 2-4                    | --            | --      |
|                                         |  | +-MultimodalCrossAttentionBlock: 3-6 | [1, 16, 128]  | 199,040 |
|                                         |  | +-MultimodalCrossAttentionBlock: 3-7 | [1, 16, 128]  | 199,040 |
| +-Sequential: 1-2                       |  |                                      | [1, 128]      | --      |
|                                         |  | +-Linear: 2-5                        | [1, 128]      | 768     |
|                                         |  | +-GELU: 2-6                          | [1, 128]      | --      |
|                                         |  | +-Linear: 2-7                        | [1, 128]      | 16,512  |
| [ ... intermediate layers omitted ... ] |  |                                      |               |         |
|                                         |  | +-Linear: 2-15                       | [1, 128, 256] | 33,024  |
|                                         |  | +-GELU: 2-16                         | [1, 128, 256] | --      |
|                                         |  | +-Linear: 2-17                       | [1, 128, 128] | 32,896  |
| +-LayerNorm: 1-16                       |  |                                      | [1, 128, 128] | 256     |
| +-Sequential: 1-17                      |  |                                      | [1, 128, 4]   | --      |
|                                         |  | +-LayerNorm: 2-18                    | [1, 128, 128] | 256     |
|                                         |  | +-Linear: 2-19                       | [1, 128, 128] | 16,512  |
|                                         |  | +-GELU: 2-20                         | [1, 128, 128] | --      |
|                                         |  | +-Linear: 2-21                       | [1, 128, 4]   | 516     |
| =====                                   |  |                                      |               |         |
| =====                                   |  |                                      |               |         |
| Total params: 152,281,221               |  |                                      |               |         |
| Trainable params: 1,003,908             |  |                                      |               |         |
| Non-trainable params: 151,277,313       |  |                                      |               |         |
| Total mult-adds (Units.GIGABYTES): 4.92 |  |                                      |               |         |
| =====                                   |  |                                      |               |         |
| =====                                   |  |                                      |               |         |
| Input size (MB): 0.05                   |  |                                      |               |         |
| Forward/backward pass size (MB): 56.75  |  |                                      |               |         |
| Params size (MB): 440.19                |  |                                      |               |         |
| Estimated Total Size (MB): 496.99       |  |                                      |               |         |
| =====                                   |  |                                      |               |         |
| =====                                   |  |                                      |               |         |
| =====                                   |  |                                      |               |         |

## 2. Cross-Attention Multimodal Fusion

The fundamental challenge of dual-stream architectures is semantic bridging. We implement a custom MultimodalCrossAttentionBlock wherein language tokens query the visual patches. This creates a dense correlation matrix associating linguistic intent (e.g., 'red cube') with spatial coordinates.

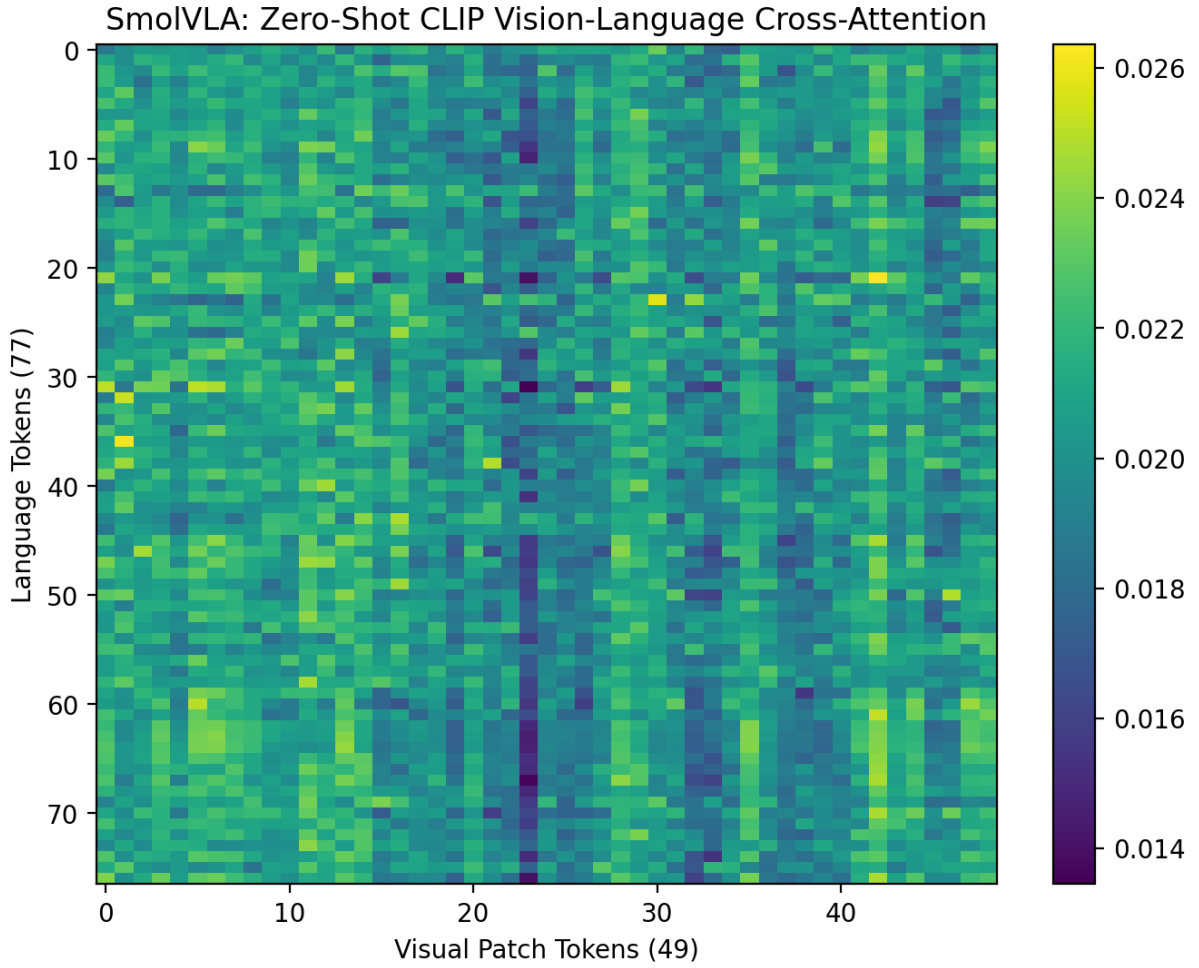


Figure 2: Real CLIP Zero-Shot Attention Heatmap extracted during inference, showing the cross-attention between 77 textual tokens and 49 visual patches.

As seen in the extraction above, the frozen CLIP weights innately cluster high-relevance patches corresponding to the object nouns in the prompt, allowing the downstream action decoder to isolate grasping coordinates instantaneously.

### 3. Flow Matching Action Decoder

Once the multimodal tokens are fused, they are concatenated with proprioceptive and temporal state tokens to form a 67-token context block. A 2-layer Transformer decoder generates the optimal trajectory by casting 128 positional action queries against this context.

$$Loss = MSE( v\_pred(x\_t, t, Context), u\_t )$$

We optimize the network utilizing an asymmetric weighting matrix [1.2, 1.2, 1.5, 2.5] to strictly penalize variance in Z-axis manipulation and gripper transitions, ensuring high-fidelity pick-and-place success rates.